FIXED-INCOME CONTINGENCY PROVISIONS

Learning Outcome

- describe common cash flow structures of fixed-income instruments and contrast cash flow contingency provisions that benefit issuers and investors

A contingency provision is a clause in a legal agreement that allows for an action if an event or circumstance occurs. The most common contingency provisions for bonds involve the right—but not the obligation—for an issuer or the bondholders to take an action specified in an indenture. These bond contingency provisions are often referred to as embedded options because they resemble option contracts but cannot be traded separately from the bond itself. The value of these embedded options may be established by comparing the value of a bond with a contingency provision with that of a bond from the same issuer without the provision. The most common bonds with contingency provisions include callable bonds, putable bonds, and convertible bonds.

Callable Bonds

An issuer of a **callable bond** has the right to redeem (or "call") all or part of the bond prior to the specified maturity date. An issuer considering callable versus non-callable bonds usually seeks the flexibility to refinance debt if market interest rates were to fall. **Fixed-price calls** grant an issuer the right to buy back the bond at a predetermined price in the future, as shown in Exhibit 9 for the Vivivyu Inc. bonds that were introduced in the prior module.

Exhibit 9:

Vivivyu (VIVU) Callable Bond Terms

6.5% Vivivyu Inc. (VIVU) Seven-Year Callable Notes (the "Notes") Prospectus Summary	
Issuer:	Vivivyu Incorporated
Settlement Date:	[T + 3 Business Days]
Maturity Date:	[Seven Years from Settlement Date]
Principal Amount:	USD400 million
Interest:	6.5% fixed coupon
Interest Payment:	Starting six months from [Settlement Date] to be paid semiannually with final payment on [Maturity Date]
Seniority:	The Notes are secured and unsubordinated obligations of VIVU and rank pari passu with all other secured and unsubordinated debt.
Call Provision:	The Issuer may redeem some or all of the Notes on any Business Day before [Maturity] starting three years after [Settlement] based on the Call Price Schedule as a percentage of the Principal Amount plus accrued interest.
Call Price Schedule:	103.25% [Three to Four Years after Settlement] 102.50% [Four to Five Years after Settlement] 101.75% [Five to Six Years after Settlement] 101.00% [Six to Seven Years after Settlement]

Callable bonds typically specify a period over which the call feature cannot be used, known as a **call protection period**. During the **call period**, the period in which the call

For example, the VIVU bond may not be called for the first three years but during the following year may be called at a price of USD103.25 per USD100 in par value. VIVU has the right to call the bond during the final four years at a fixed price that declines over time. Investors face reinvestment risk starting in year three when the bond can be called, as well as limited bond price appreciation beyond the fixed call price.

The value of the call feature depends on the relationship between the YTM and the coupon of the bond. If the YTM rises above the coupon rate, there is little incentive for the issuer to use the call feature. The YTM provides an estimate of refinancing costs, which in this case exceeds the existing coupon. Therefore, the callable bond behaves much like a non-callable equivalent bond with respect to price movements due to interest rate changes. However, as a callable bond's YTM falls below its coupon rate, its price is effectively capped at the fixed call price, while the price of a non-callable bond will continue to rise. Investors expect to be compensated on callable bonds with a higher yield versus a similar non-callable bond given the uncertain maturity and limited price appreciation, known as **call risk**.

Some bonds have a call feature, different from the fixed-price call, under which the issuer compensates investors to buy bonds back known as a make-whole call. Make-whole calls are most common among highly rated bond issues and offer issuers greater flexibility to repurchase debt than purchases in the market. However, this provision usually requires an issuer to pay bondholders a high price based on the YTM of a sovereign bond of similar maturity. Because this is usually well above the bond's current market price, make-whole calls are rarely executed and have little economic impact on issuers or investors.

Putable Bonds

A put provision gives bondholders the right to sell the bond back to the issuer at a pre-determined price on specified dates. The put price is usually the par value of the bond. **Putable bonds** benefit bondholders by guaranteeing a pre-specified selling price at the redemption dates. If interest rates rise after issuance and bond prices fall, bondholders can sell or "put" the bond back to the issuer and reinvest the proceeds at higher market interest rates. The issuer will then be forced to refinance its debt earlier than planned at higher yields.

Since a put provision has value to bondholders, the price of a putable bond will be above the price of an otherwise similar non-putable bond. The yield on a bond with a put provision must therefore be lower than the yield on an otherwise similar non-putable bond. The lower yield compensates the issuer for the value of the put option to the investor.

If a putable bond's YTM is less than its coupon rate (so the price is higher than the put price), the put feature is less valuable, and the putable bond behaves like an option-free equivalent bond with respect to interest rate changes. However, if the putable bond's YTM increases above its coupon rate, the put price serves as a price floor for the bond.

Convertible Bonds

A **convertible bond** is a debt instrument with a contingency provision related to the issuer's outstanding common equity. It grants bondholders the right to exchange the issuer's bond for a number of its common shares in the future at an effective price per share known as the **conversion price**. The conversion feature rises in value to investors as the share price appreciates. Rather than a right to retire debt early in the case of callable and putable bonds, if exercised, the convertible feature replaces debt with equity.

Exhibit 10 shows a convertible bond for ZTG BioTech S.p.a., a publicly traded Italian company.

Exhibit 10: ZTG BioTech S.p.a. Convertible Bond Terms

Issuer:	ZTG Biotech S.p.a.
Settlement Date:	[T + 3 Business Days]
Maturity Date:	Five Years from Settlement Date unless the Notes are redeemed earlier in a Conversion
Principal Amount:	EUR300 million
Interest:	1.25% fixed coupon to be paid annually starting One Year from the Settlement Date with the final coupon paid at Maturity
Seniority:	The Notes are an unsecured obligation of ZTGB and rank pari passu with other unsecured debt
Conversion Provision:	An Investor has, at its sole option, the right to convert a portion or the full sum of the Principal Amount, or any part outstanding at any time at the Conversion Price during the Conversion Period
Current Share Price:	EUR28.00 per ZTGB common equity share
Conversion Price:	EUR42.00 per ZTGB common equity share
Conversion Period:	Any Business Day starting One Year from the Settlement Date through the Maturity Date

The 1.25% annual coupon on the ZTGB convertible bond is well below the YTM investors would expect for a standard ZTGB bond of the same maturity. Bondholders are often willing to accept a yield-to-maturity that is very low (or even zero) in exchange for conversion rights. ZTGB bond investors have the right to exchange debt for ZTGB common stock at a price of EUR42.00 per share, which represents a 50% premium over the current ZTGB price of EUR28.00 per share [= (42.00 − 28.00)/28.00]. The **conversion ratio** represents the number of common shares a bond may be converted into for a specific par value:

$$\text{Conversion ratio} = \text{Convertible bond par value}/\text{Conversion price}. \quad (2)$$

For example, if the ZTGB bond is traded in EUR1,000 face value units,

- the conversion ratio is equal to EUR1,000/EUR42, or 23.81, and
- an owner of one EUR1,000 unit of the ZTGB bond has the right to redeem the bond for 23.81 shares of ZTGB stock during the conversion period.

One way to estimate the value of the conversion feature at any time is to compare the convertible bond's price with its value if the bondholder were to convert the bonds today. This **conversion value** is calculated as follows:

$$\text{Conversion value} = \text{Conversion ratio} \times \text{Current share price}. \quad (3)$$

For example, assume that one year from ZTGB's issuance date, the issuer's shares are trading at EUR35.00. A ZTGB bondholder could therefore convert one EUR1,000 unit to a conversion value of EUR833.35 (= 23.81 × EUR35.00). If the convertible bond is trading at par (or a market price of EUR1,000 per unit), the investor will prefer to hold the bond and receive interest and principal rather than convert the bond to shares. In general, a convertible bond's price will vary with that of the issuer's shares. If the share price is well below the conversion price, the convertible bond's price will approach that of a standard non-convertible bond. If the issuer's share price far exceeds the conversion price, the bond's price will more closely track its conversion value.

The use of convertible debt is common among growth companies, which may have limited cash flow to pay interest and repay principal but are willing to raise equity at a higher conversion price in the future. Convertible bonds also often include additional features, such as a call provision, allowing issuers to limit bond investor gains from share price appreciation by redeeming bonds early.

Some issuers may issue warrants with bonds. A **warrant** is an "attached," rather than embedded, option entitling the bondholder to buy the issuer's stock at a fixed exercise

the Hong Kong Stock Exchange.

Several European banks have issued a type of convertible bond called **contingent convertible bonds** (CoCos), with contingent write-down provisions. While standard convertibles will be converted when the share price increases, CoCos are converted on the downside. This conversion is also not at the discretion of the investor but happens automatically if a specified event occurs—for example, if the bank's core Tier 1 capital ratio (a measure of the bank's proportion of core equity capital available to absorb losses) falls below a minimum regulatory requirement. If the bank experiences losses that reduce its equity capital below the minimum regulatory capital requirement, CoCos convert debt into equity to increase its capital and reduce the bank's likelihood of default. CoCos were introduced in order to limit systemic risk, or the risk of financial system failure. For this reason, CoCos offer investors a higher yield than otherwise similar bonds. Note that the conversion is triggered by a specific event, the breach of certain minimum regulatory capital requirements, and not by certain equity or debt price levels.

Question Set

1. All else equal, the bond with the greatest benefit to the issuer is a bond with:

A. no contingency provisions.
B. a fixed-price call provision.
C. a make-whole call provision.

Hide Solution

Solution:

B is correct. Fixed-price calls grant the issuer the right to buy back the bond at a predetermined price in the future, which is a benefit to the issuer if interest rates fall.

A is incorrect. A bond with no contingency provisions is an option-free bond and is not as beneficial to the issuer as a bond with a fixed-price call provision.

C is incorrect. A make-whole call is a contingency feature under which issuers can buy bonds back at a price usually based on the yield-to-maturity of a sovereign bond of similar maturity plus a predetermined spread. Make-whole calls are rarely executed and have little economic impact on issuers or investors.

2. Callable and putable bond prices with the same coupon rate will behave most like an option-free equivalent bond when their YTMs are, respectively:

A. above the coupon rate and below the coupon rate.
B. below the coupon rate and above the coupon rate.
C. both near the coupon rate.

Hide Solution

Solution:

A is correct. If a callable bond's YTM rises above its coupon rate, the call feature is less valuable and a callable bond behaves much like a non-callable bond with respect to price movements due to interest rate changes. If a putable bond's YTM is less than its coupon rate, the put feature is less valuable and the putable bond's price behaves like an option-free equivalent bond with respect to interest rate changes.

B is incorrect. This is the reverse of the correct relationship.

price changes would be lower in magnitude relative to a non-callable bond.

3. A convertible bond will closely track its conversion value if the issuer's share price is:

A. below the conversion price.
B. equal to the conversion price.
C. above the conversion price.

Hide Solution

Solution:

C is correct. In general, a convertible bond's price will vary with that of the issuer's shares. If the issuer's share price far exceeds the conversion price, the bond's price will more closely track its conversion value.

A is incorrect. If the share price is well below the conversion price, the convertible bond's price will approach that of a non-convertible bond.

B is incorrect. In this case, the convertible bond would have a mix of price properties between that of a non-convertible bond and a convertible bond.

4. Match each description in the right column with the correct bond in the left column.

A. Convertible bond	I. Conversion occurs automatically
B. Contingent convertible (CoCo) bond	II. Conversion occurs at the discretion of the investor
	III. More likely to convert if share price increases
	IV. More likely to convert if share price decreases

Hide Solution

Solution:

A. II and III are correct. A standard convertible bond grants the right to redeem a bond for an issuer's equity, occurs at the discretion of the investor, and is more likely if the issuer's share price increases.
B. I and IV are correct. Contingent convertible (CoCo) bonds are convertible on the downside and occur automatically if a specified event occurs, such as a decline in bank capital below a regulatory minimum level.

5. An issuer that would like to conserve near-term cash outflows would most likely issue a:

A. putable bond.
B. callable bond.
C. convertible bond.

Hide Solution

Solution:

C is correct. All else equal, an issuer would have the lowest interest payments on a convertible bond. Bondholders are often willing to accept a yield-to-maturity that is very low (or even zero) in exchange for conversion rights. Putable bonds offer lower yields than callable bonds because the put option is a benefit to the bondholder while the call option is a benefit to the issuer. However, the yields on callable and putable bonds are high compared to yields on convertibles.

Discussion

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